

Pentagon Robot (5-Bar Parallel Arm Pick and Place Automation)

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Abstract

The Pentagon Robot project was developed to address the need for an accessible educational platform that demonstrates advanced robotic kinematics through real hardware. The system is a five-bar parallel planar linkage designed for pick-and-place operations, capable of transferring spherical objects between matrix locations using randomized motion control. In doing so, it connects theoretical kinematics with practical robotic implementation.

A Model-Based Design approach was used throughout the project, combining analytical kinematics, CAD modeling, simulation, and iterative prototyping. The mechanical design uses two base-mounted NEMA 17 stepper motors in a stacked direct-drive configuration to reduce moving inertia, prevent link interference, and maximize workspace. After the robot was modeled in SolidWorks, the design was exported as a URDF and integrated into ROS 2 Humble. The software architecture included nodes for suction hardware control, inverse kinematics, trajectory planning, and pick-and-place planning, all launched through a unified system and visualized in RViz2.

Simulation testing was first conducted to verify workspace reachability and avoid singularity regions before deployment on physical hardware. After integration, further tuning improved repeatability and overall performance. The final prototype met the project's functional goals within a \$350 budget, making it a practical and scalable robotics education platform.

Introduction/Problem statement

Modern manufacturing relies heavily on automated pick-and-place systems to achieve high levels of speed, precision, and efficiency. While the underlying principles of these systems are well established in theory, there remains a significant gap in accessible platforms that allow students to interact with and understand advanced robotic mechanisms in practice. Most educational robotic systems are limited to serial manipulators or Cartesian gantries, which, while effective for basic learning, do not expose users to the complexities and advantages of parallel kinematic systems. In contrast, industrial applications increasingly utilize mechanisms such as five-bar linkages due to their superior rigidity, higher speed capabilities, and improved dynamic performance.

This gap limits hands-on learning opportunities for closed-loop kinematic systems, which are essential for understanding non-linear motion, workspace constraints, and advanced control strategies. The Pentagon Robot project addresses this challenge by developing a low-cost, open-source platform that bridges the divide between theoretical concepts and real-world implementation. By integrating mechanical design with algorithmic control, the system enables users to explore advanced robotics concepts without the high cost and complexity associated with industrial-grade systems.

The primary objective of this project is to design, simulate, and fabricate a five-bar parallel planar robot capable of performing automated pick-and-place operations. The system is designed to transfer ping-pong balls between predefined matrix locations using a randomized control algorithm, with a target cycle time of 10 seconds or less and a placement accuracy of ± 0.5 mm. A hybrid control architecture, utilizing a Raspberry Pi and Arduino UNO R3, ensures effective coordination between high-level planning and low-level actuation, all within a constrained budget of \$350.

Decision Matrices

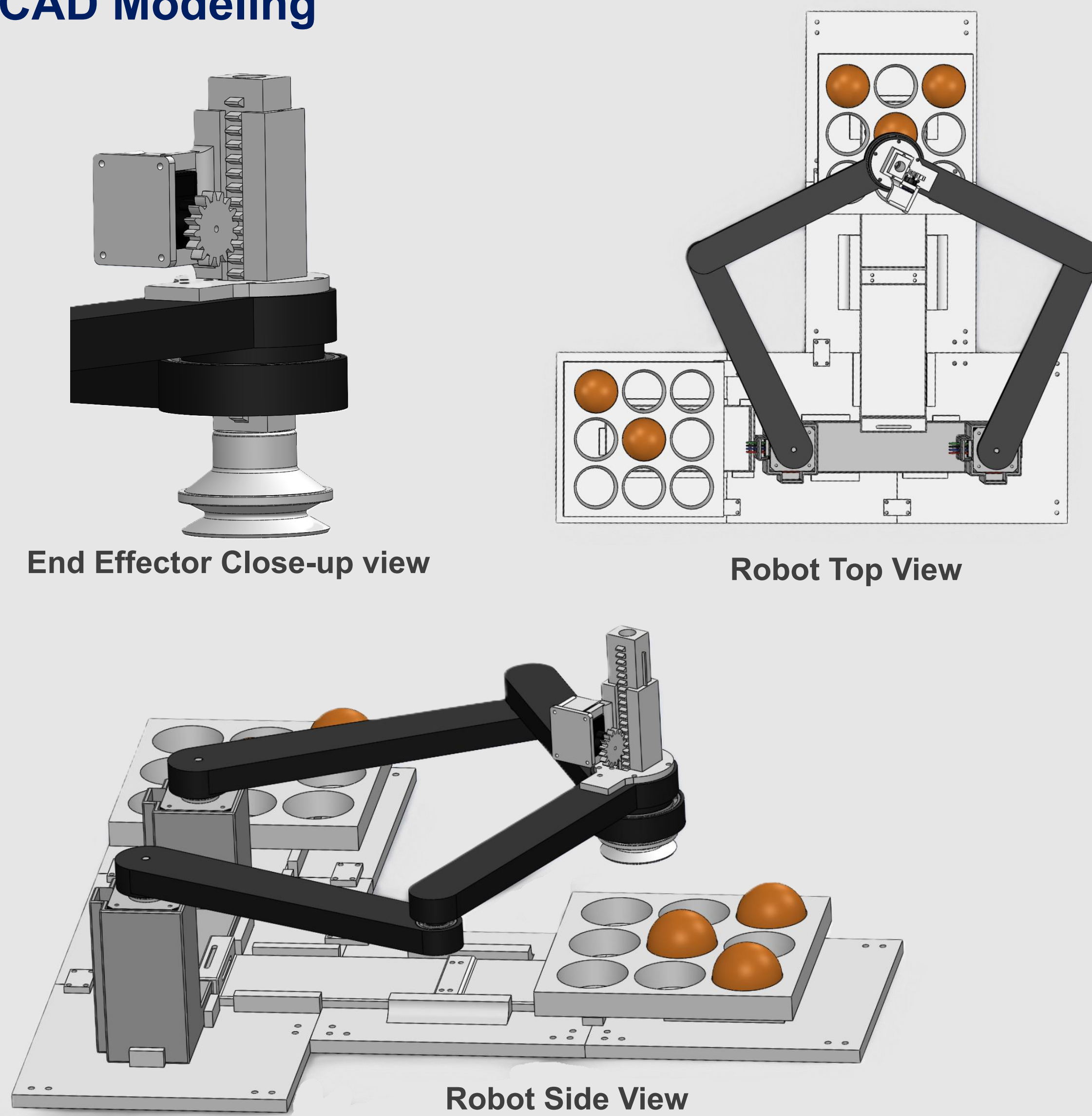
Design decision matrices were used to systematically evaluate multiple design concepts and ensure the most suitable solution was selected. Key criteria such as performance, complexity, cost, and reliability were weighted based on project requirements, and each concept was scored accordingly. This structured approach minimized bias, enabled objective comparison, and ensured that the final design choices, such as the stacked-motor configuration and vacuum end effector, provided the best overall balance of functionality, efficiency, and feasibility for the application.

Note: 1–5 rating scale, where 5 is best.

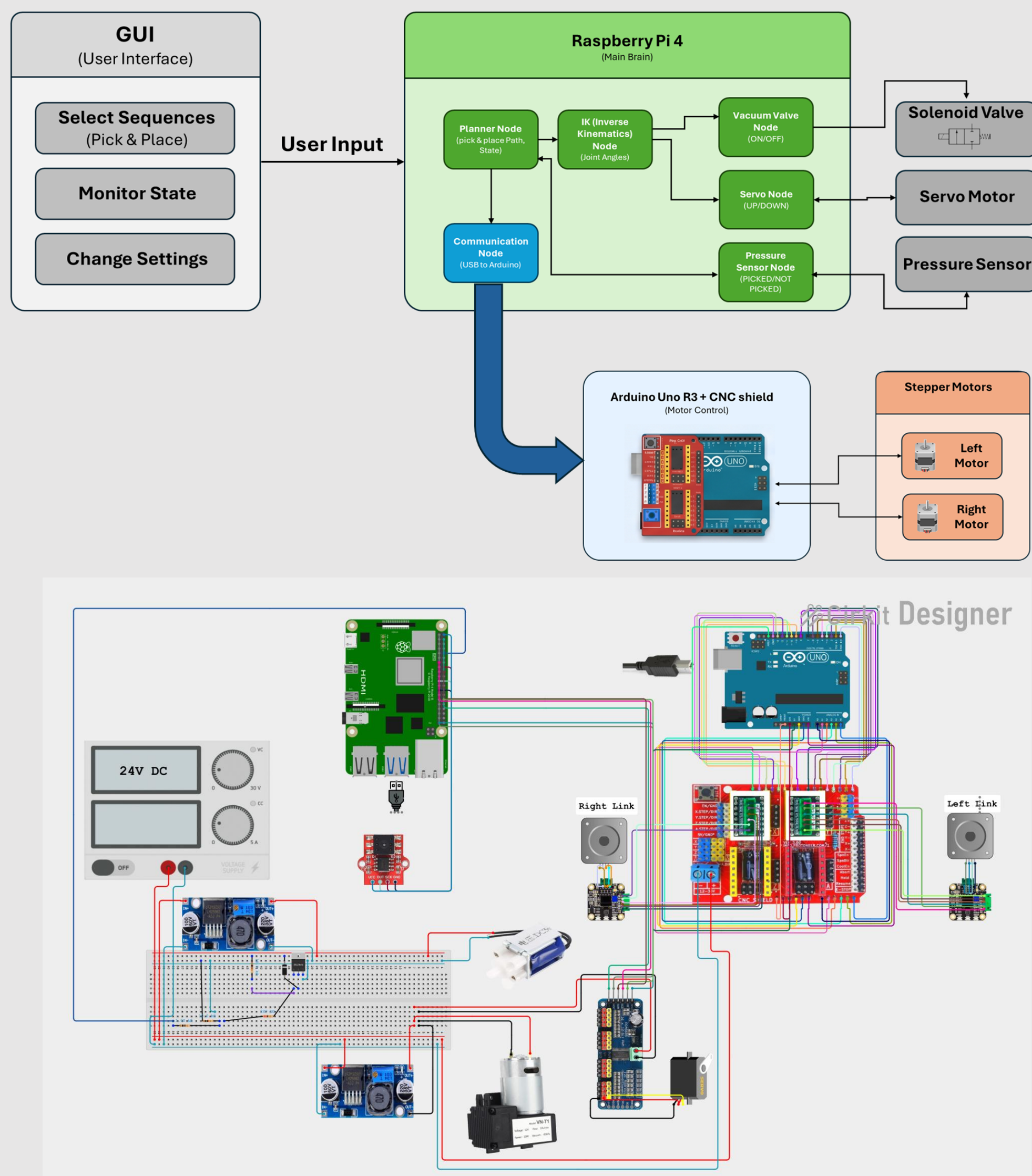
Motor Configuration					
Criteria	Weight (%)	Stacked Base-Mounted Direct Drive	Side-by-Side Base-Mounted Direct Drive	Belt-Driven Remote Actuation	Overhead Motor-Mounted
Low moving inertia	20	5	4	5	2
Workspace utilization	20	5	4	4	3
Link interference avoidance	15	5	3	4	3
Mechanical simplicity	15	4	4	2	3
Control simplicity	10	5	5	3	4
Manufacturability / assembly	10	4	4	2	3
Cost	10	4	4	2	3
Total Weighted Score	100	4.7	3.95	3.45	2.85

End effector					
Criteria	Weight (%)	Vacuum Suction Cup	Mechanical Gripper	Soft Two-Finger Gripper	Magnetic Gripper
Ability to handle spherical objects	20	5	4	4	1
Pick-and-place speed	20	5	3	3	5
End effector weight	15	5	3	3	4
Mechanical simplicity	15	4	3	3	5
Control simplicity	10	4	3	3	5
Reliability for target object	10	4	4	4	1
Cost	10	4	3	2	3
Total Weighted Score	100	4.55	3.35	3.2	3.25

CAD Modeling



Control System Logic & Circuit Diagram



Methodology

The development of the Pentagon Robot followed a structured and iterative engineering process that integrated concept selection, mechanical design, and control implementation. Multiple design configurations were first evaluated using weighted decision matrices based on criteria such as moving inertia, workspace efficiency, interference avoidance, simplicity, and cost, leading to the selection of a stacked base-mounted five-bar linkage with a vacuum suction end effector. The mechanical system was then designed and optimized in CAD to ensure proper link alignment, rigidity, and minimal backlash, while standard, low-cost components were selected to maintain accessibility and reproducibility.

In parallel, a modular software architecture was developed using ROS 2, where separate nodes handled inverse kinematics, trajectory planning, pick-and-place sequencing, suction control, servo actuation, and pressure sensing. A Raspberry Pi 4 was used for high-level computation and coordination, while an Arduino UNO R3 with a CNC shield managed low-level motor control. A user interface was implemented to allow real-time selection of pick-and-place operations and system parameters. During operation, a pressure sensor provides feedback to confirm successful object pickup, enabling a closed-loop control strategy that automatically retries failed attempts. The system was iteratively tested and calibrated in both simulation and hardware to refine motion accuracy, reliability, and cycle time within the project constraints.

Results

The final prototype successfully achieved the project's performance objectives and demonstrated reliable automated operation. Experimental testing showed consistent repeatability across multiple trials, with a high success rate in object pickup and placement using the vacuum-based end effector. The integrated feedback loop, based on pressure sensing, improved reliability by allowing the system to detect and recover from unsuccessful grasp attempts. The system met all functional and cost constraints, demonstrating an effective and scalable solution for educational pick-and-place robotics.

Performance Matrix

Metric	Target	Achieved
Cycle Time	≤ 10 s	~ 9.6 s
Accuracy	± 0.5 mm	$\sim \pm 0.4$ mm
Success Rate	—	$\sim 90\%$
Budget	\$350	$\sim \$340$

Conclusion

The Pentagon Robot project demonstrates that advanced parallel robotic systems can be designed and implemented using accessible components without compromising functionality. Through the integration of mechanical design, control systems, and real-time feedback, the project highlights the importance of combining analytical modeling with practical engineering considerations such as manufacturability, cost, and reliability. The use of a modular architecture and open-source tools makes the system adaptable for further development and educational use. Key insights from this work include the effectiveness of closed-loop feedback in improving system robustness and repeatability while in applications. While the current system meets its functional objectives, future improvements could focus on enhancing grasp reliability, refining trajectory optimization, and expanding the system for more complex tasks by adding a vision capability. Overall, the project serves as a practical demonstration of how theoretical concepts in kinematics and control can be successfully translated into a working robotic system.

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